Trial ordering in repeated exposure virtual reality pedestrian research: comparing fixed and participant specific randomised sequences

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ABSTRACT

When every participant receives the same trial sequence, each condition is tied to one trial position and preceding experience. We compared two non concurrent runs of a virtual reality pedestrian interaction experiment. Participants were not assigned at random to study run. Fifty participants received participant specific randomised sequences and 50 received one fixed sequence of the same 40 trials. The main measure was the percentage of 100 ms prepassage intervals in which a trigger indicated perceived crossing unsafety. The estimate was 40.48% in the randomised run and 42.82% in the fixed run, a difference of 2.34 percentage points (95% CI $[-5.16, 9.83]$, $p = 0.54$). The confidence interval did not establish equivalence. No reliable overall differences were detected for additional trigger or head movement measures, although condition response profiles differed for several outcomes. The strongest condition specific contrast concerned the perceived influence of pedestrian spacing in a yielding, eHMI off, participant second, 2 m condition. The fixed run was 23.34 rating points lower (globally adjusted $p = 0.011$), and this condition always occurred first in the fixed sequence. Its effect could not be separated from early session experience or sample differences. In the randomised run, the estimated eHMI contribution to reported vehicle intention understanding increased with trial position and prior relevant exposure, but these changes did not differ reliably between runs. Confounding of scheduling with study run precludes causal attribution, but the comparison demonstrates why fixed sequences weaken the interpretation of condition specific and experience related effects.

1. Introduction

Repeated measures designs are widely used in traffic psychology because they allow the same participants to experience several experimental conditions and serve as their own reference. This reduces the influence of stable individual differences and makes it possible to examine multiple aspects of a traffic situation within a manageable sample (Greenwald, 1976; Charness, Gneezy and Kuhn, 2012). Such designs are common in pedestrian research, where participants may encounter different vehicle movements, communication signals, road configurations, and arrangements of other road users (Camara, Dickinson and Fox, 2021; Feng, Duives, Daamen and Hoogendoorn, 2021; Tran, Parker and Tomitsch, 2021). Immersive virtual reality is particularly useful for this research because potentially safety critical interactions can be presented under controlled and repeatable conditions without exposing participants to physical traffic (Camara et al., 2021; Tran et al., 2021). However, responses in repeated measures experiments may depend not only on the condition being presented, but also on when it is presented and what the participant has previously experienced.

This creates an important interpretive problem when all participants receive the same fixed trial sequence. In a fixed sequence, each condition always occurs at the same stage of the experiment and after the same preceding trials. A condition presented early may be encountered while participants are still becoming familiar with the task or learning how to use the response scale. A condition presented later may be judged after participants have developed expectations about the experimental situations or adjusted their response strategy. An observed difference between conditions may therefore reflect the intended manipulation, differences in accumulated experience, or both. Trial ordering is consequently not merely a procedural choice, but an important part of how condition effects should be interpreted.

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Trial ordering can also affect conclusions about changes over the course of an experiment. Responses may change because participants learn the meaning of a signal, become familiar with the task, develop expectations, calibrate their judgements, or adopt a different response strategy. However, when different types of conditions are concentrated in the early and late parts of a fixed sequence, an apparent change across trials may instead reflect differences in the conditions being presented. Condition effects and experience related changes can therefore become confounded. Participant specific randomisation reduces this association by distributing conditions across different trial positions and preceding experiences (Fisher, 1966; Montgomery, 2017). It does not prevent learning or familiarisation, but makes these processes less consistently tied to particular conditions. Counterbalancing, Latin square arrangements, and balanced sequence designs provide related alternatives when unrestricted randomisation could produce undesirable repetitions or implausible transitions (Williams, 1949; Jones and Kenward, 2003; Brooks, 2012).

These concerns are especially relevant to pedestrian interactions with automated vehicles. Pedestrians judge whether an approaching vehicle intends to yield by combining information from its speed, distance, and deceleration with information from the surrounding traffic situation (Rasouli and Tsotsos, 2019; De Clercq, Dietrich, Núñez Velasco, De Winter and Happee, 2019; Dietrich, Tondera and Bengler, 2020). External human machine interfaces, or eHMIs, have been proposed as an additional means of communicating the vehicle's status, awareness, or intended behaviour (Habibovic, Lundgren, Andersson, Klingegård, Lagström, Sirkka, Fagerlönn, Edgren, Fredriksson and Krupenia, 2018; Dey, Habibovic, Löcken, Wintersberger, Pfleging, Riener, Martens and Terken, 2020). Previous research indicates that eHMIs can support pedestrians' understanding of vehicle intentions and their crossing judgements under some conditions (De Clercq et al., 2019; Dey, Van Vastenhoven, Cuijpers, Martens and Pfleging, 2021; **?**). Their contribution nevertheless depends on the design of the interface, the traffic context, and the consistency between the communicated message and the movement of the vehicle. When yielding behaviour is already clear from vehicle kinematics, an eHMI may provide little additional information (Dietrich et al., 2020; Dey et al., 2021; de Winter and Dodou, 2022). When vehicle intention is less clear, the eHMI may become more influential, although unfamiliar, ambiguous, or inconsistent signals may also be misunderstood.

Repeated encounters may further change how pedestrians interpret and use an eHMI. Familiarity with an eHMI has been associated with changes in its interpretation, perceived safety, trust, and influence on crossing judgements (Lee, Madigan, Uzondu, Garcia, Romano, Markkula and Merat, 2022; Colley, Bajrovic and Rukzio, 2022; Colley, Kornmüller, Dey, Ju and Rukzio, 2024). Repeated exposure to automated vehicle interactions has also been associated with changes in trust, crossing initiation, visual behaviour, and psychophysiological responses (Hübner, Stockmann and Bengler, 2025; Shahi, Kumar Debnath, Birrell, Horan, Alatrash and Payre, 2026). These changes may reflect learning the association between the eHMI and vehicle behaviour, increasing confidence in the signal, or greater reliance on it. Greater reliance is not necessarily beneficial, particularly when the eHMI conflicts with the vehicle's movement or with other information in the traffic environment (Kaleefathullah, Merat, Lee, Eisma, Madigan, Garcia and De Winter, 2022; Hensch, Kreißig, Beggiato and Krems, 2022). Moreover, a numerical trend across trials should not automatically be interpreted as learning because similar trends may also result from task familiarisation, changing expectations, scale calibration, or adjustments in response strategy.

Previous pedestrian research has examined vehicle kinematics, eHMI design, familiarity, trust, and behavioural adaptation across repeated encounters (De Clercq et al., 2019; Dey et al., 2021; Lee et al., 2022; Colley et al., 2022, 2024; Hübner et al., 2025; Shahi et al., 2026). Less attention has been given to whether conclusions remain consistent when the same experimental conditions are presented using a fixed sequence rather than participant specific randomised ordering. Similar average responses would not necessarily mean that trial order is unimportant, because the scheduling approaches could produce different patterns across individual conditions or different interpretations of changes with experience. The present study therefore examines trial ordering at three levels: average responses across conditions, the pattern of responses to individual conditions, and changes associated with experience during the experimental session.

1.1. Aim of the study

The aim of this study was to examine whether conclusions from a repeated exposure virtual reality pedestrian experiment were consistent across two independent study runs that differed in trial scheduling. We compared one fixed sequence with participant specific randomised sequences for three purposes: first, to compare average responses across the complete set of conditions; second, to examine whether the response pattern across conditions differed between runs; and third, to examine whether the runs supported different conclusions about change with experience. Particular attention was given to perceived crossing safety and reported understanding of the external human machine interface, because both may develop as participants encounter similar traffic situations repeatedly. Because study run and trial scheduling method were not independently randomised, between run contrasts were interpreted as comparisons between the two samples rather than as causal effects of ordering method.

2. Method

2.1. Study design

The study was a non concurrent comparison of two independent runs of the same virtual reality pedestrian experiment. Both runs used the same apparatus, virtual environment, task, procedure, and 40 experimental conditions. In the fixed run, all participants completed the conditions in one shared sequence. In the randomised run, the conditions were shuffled separately for each participant. The fixed run was conducted before the randomised run. Trial scheduling method was therefore completely associated with study run, sample, and time of data collection. The design provides a methodological case comparison of the conclusions obtained under the two schedules, not a randomised estimate of the effect of trial ordering.

Each run included 50 participants, and each participant completed all 40 main trials. The analysis therefore included 100 participants and 4,000 participant trial observations. The factorial effects within the randomised run are reported separately by Alam, Dey, Martens and Bazilinskyy (2026); that manuscript concerns the effects of the traffic manipulations, whereas the present study concerns the robustness and interpretation of conclusions under different trial schedules.

2.2. Participants

The study received approval from the Ethics Review Board of Eindhoven University of Technology. All participants provided informed consent before taking part.

Participants in the randomised run had a mean age of 28.30 years ($SD = 7.30$); 28 identified as male, 21 as female, and one selected another or undisclosed category. Participants in the fixed run had a mean age of 25.28 years ($SD = 4.61$); 29 identified as male, 20 as female, and one selected another or undisclosed category. Recent experience with virtual reality was also recorded. Because the two study runs were conducted at different times and the fixed sample had a lower mean age, age, gender, and recent virtual reality experience were included in a sensitivity analysis.

2.3. Hardware

Both runs used the same immersive laboratory setup. Participants wore a Meta Quest 3 head mounted display and held a handheld controller. The virtual environment was implemented in Unity 2022.3.5f1 and was adapted from the coupled automated vehicle–pedestrian simulator described by Dey et al. (2021). The analogue controller trigger and head mounted display position and orientation were recorded at 50 Hz.

2.4. Virtual environment and vehicle behaviour

The virtual environment depicted a straight residential street without a formal pedestrian crossing. Participants stood stationary at the kerb and were asked to imagine that they intended to cross the road. A stationary virtual pedestrian was positioned on the same pavement. An automated vehicle approached from the participant's left.

In non yielding trials, the vehicle maintained a speed of approximately 50 $\mathrm{km/h}$. In yielding trials, it began decelerating at 2.4 $\mathrm{m/s^2}$ from a position 43 m before the first pedestrian, stopped 3 m before that pedestrian for 3 s, and then resumed driving.

A cyan light bar on the front of the vehicle served as the external human machine interface. In the eHMI off condition, the light bar remained inactive. In the eHMI on condition, it showed a steady approach state while the vehicle approached and an inward wiping animation when the vehicle yielded. The vehicle movement and eHMI design were informed by previous studies of automated vehicle interaction with pedestrians (De Clercq et al., 2019; Habibovic et al., 2018; Rasouli and Tsotsos, 2019).

2.5. Experimental design

The study used a $(2 \times 2 \times 2 \times 5)$ repeated measures design. The four factors were:

- vehicle behaviour: yielding or non yielding;
- eHMI status: off or on;

- relative pedestrian order: participant first or participant second in the vehicle's path; and
- centre to centre spacing between the participant and the other pedestrian: 2, 4, 6, 8, or 10 m.

Each participant completed one trial for every combination of these factors, resulting in 40 main trials.

The relative pedestrian order manipulation also changed the spatial arrangement of the scene. When the participant was first, the participant occupied the first roadside position encountered by the approaching vehicle. When the participant was second, the stationary pedestrian occupied the first position and the participant stood farther along the vehicle's path. The vehicle's yielding location was defined relative to the first pedestrian position. Relative pedestrian order should therefore be interpreted as a combined manipulation of order and spatial geometry rather than as a manipulation of visibility alone.

2.5.1. Trial ordering

In the randomised run, the 40 experimental conditions were shuffled separately for each participant using the trial ordering procedure implemented in Unity C#. The realised trial order was reconstructed from the recorded response files and matched to the corresponding participant condition mapping. In the fixed run, every participant received the same predetermined sequence.

The realised order was retained for every participant and used in the analyses of trial position and prior exposure. The randomised run contained 49 unique realised sequences because two participants received the same sequence. Sensitivity analyses therefore examined whether excluding both participants changed the main conclusion.

Figure 1 shows how the experimental factors were distributed over trial position. In the randomised run, each factor level was distributed across different stages of the session when averaged across participants. In the fixed run, each condition was tied to one particular trial position.

2.6. Procedure

Participants first read an information sheet, provided informed consent, and completed the demographic questionnaire. They then received an explanation and demonstration of the controller trigger.

Participants were instructed to hold the trigger whenever they considered it unsafe to initiate a road crossing and to release it whenever they considered it safe. They could press and release the trigger more than once during a trial. Participants indicated their perceived crossing safety but did not physically cross the virtual road.

Two practice trials were completed before the 40 main trials. Optional breaks were offered after trials 14 and 26. After every main trial, participants answered three questions using sliders ranging from 0 to 100.

2.7. Dependent measures

The main dependent measures were perceived crossing safety and three post trial ratings. Perceived crossing safety was derived from the controller trigger. The ratings measured the perceived influence of the other pedestrian, the perceived influence of the spacing between the pedestrians, and understanding of the vehicle's intention. We also analysed additional measures of trigger use and horizontal head movement to examine whether similar average responses could conceal differences in how participants responded.

2.7.1. Perceived crossing safety

Perceived crossing safety was measured during the five seconds immediately before the vehicle passed the participant. This common window provided the same measurement duration in yielding and non yielding trials and captured the period in which the participant made the immediate crossing judgement.

The five second window was divided into 50 non overlapping intervals of 100 ms. An interval was classified as unsafe when at least one valid trigger sample exceeded 0.10. For each trial, perceived crossing unsafety was calculated as the percentage of the 50 intervals classified as unsafe. A higher percentage therefore indicated that the participant considered initiating a crossing unsafe for a greater proportion of the prepassage period.

2.7.2. Post trial ratings

After each trial, participants answered the following questions:

Q1 "On a scale of 0 to 100, how much did the behaviour of the other pedestrian influence your decision to cross the road?"

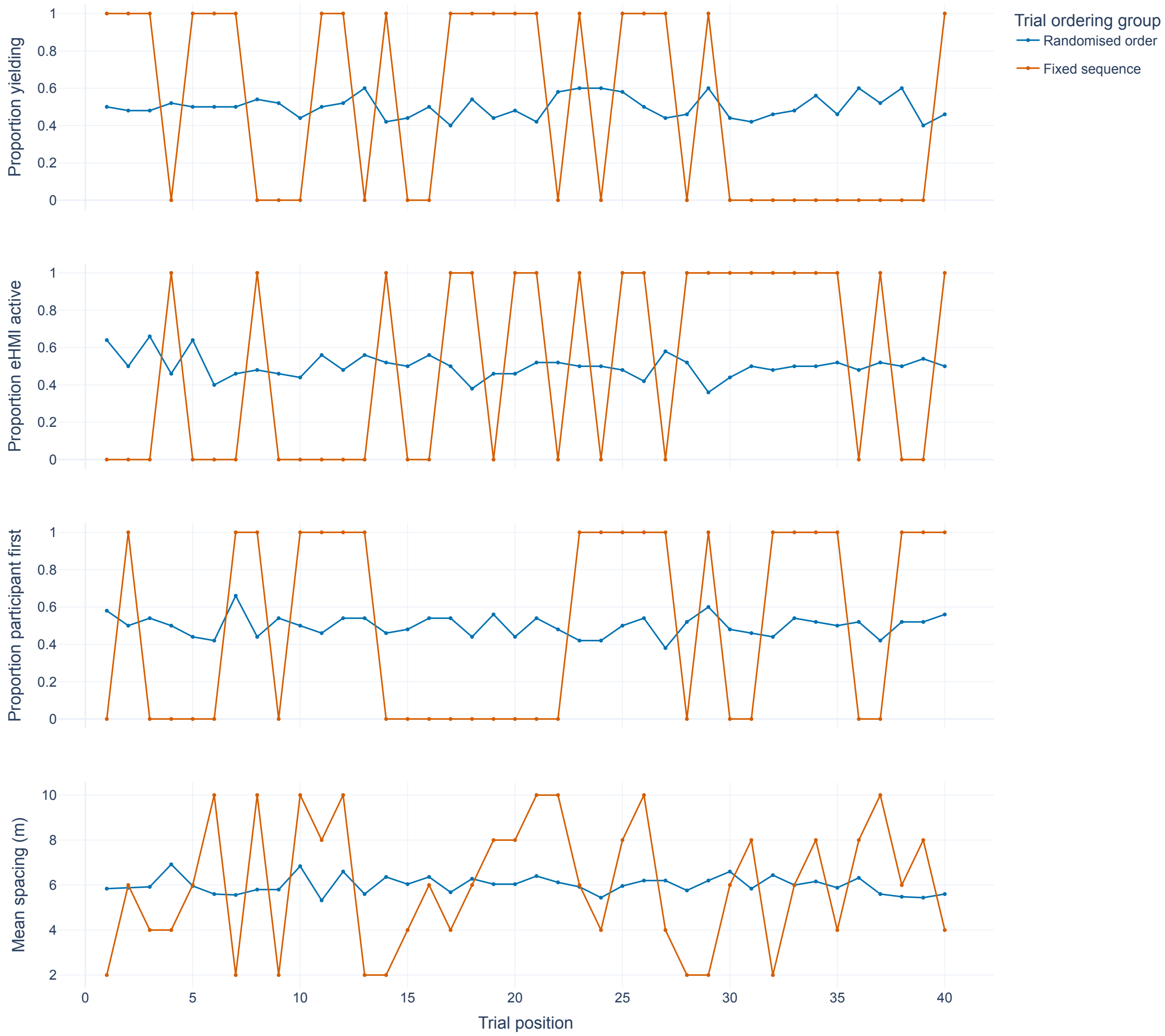

Figure 1: Distribution of the experimental factors across realised trial position. Values for the randomised run represent proportions or means across participants. Values for the fixed run represent the one sequence completed by all participants.

Q2 "On a scale of 0 to 100, how much did the distance between you and the other pedestrian affect your decision to cross the road?"

Q3 "On a scale of 0 to 100, rate how well you understood the intention of the vehicle."

Q1 measured the perceived influence of the other pedestrian's behaviour, Q2 measured the perceived influence of pedestrian spacing, and Q3 measured understanding of the vehicle's intention. The full meanings of the questions are repeated where necessary in the Results to avoid relying on the labels Q1, Q2, and Q3 alone.

2.7.3. Additional behavioural measures

Additional trigger measures described how participants used the controller. These included the mean and peak analogue trigger values, whether the trigger was activated at least once, the time of the first activation, whether the trigger was subsequently released, and the time of the first release after activation.

Horizontal head movement was analysed as an additional indication of orientation towards the approaching vehicle. Head mounted display quaternions were converted to horizontal heading by rotating Unity's forward vector and projecting it onto the horizontal plane. When multiple orientation samples occurred at the same timestamp, the quaternions were averaged using the eigenvalue method described by Markley, Cheng, Crassidis and Oshman (2007). The resulting trajectories were unwrapped and centred relative to the mean heading recorded from 0.02 to 0.30 s after trial onset. Negative heading values represented movement towards the left, where the vehicle approached.

The principal head movement measure was the mean baseline corrected heading during the 200 ms interval centred on vehicle passage. Additional measures described the minimum heading before passage, the change in heading around passage, variability within the prepassage window, and mean absolute yaw rate. Head mounted display orientation was treated as a measure of head movement and not as a direct measure of gaze or visual attention.

2.8. Statistical analysis

All trial level analyses accounted for the 40 repeated observations contributed by each participant. The analyses were organised according to their psychological purpose: comparing average responses, examining whether the experimental conditions were interpreted differently, and examining changes associated with experience. Perceived crossing unsafety and its overall between run contrast constituted the primary outcome and comparison. Post trial ratings, additional trigger measures, and head movement measures were secondary outcomes. Broader prior exposure models and flexible temporal models were exploratory. This hierarchy guided the interpretation of multiplicity adjusted results. The direction of every reported between run contrast was fixed run minus randomised run.

Perceived crossing unsafety was analysed using the numbers of unsafe and safe 100 ms intervals in each trial. These counts were entered into a grouped binomial model with participant clustered robust standard errors. Continuous ratings, trigger measures, and head movement measures were analysed using Gaussian generalised estimating equations. Binary trigger outcomes were analysed using binomial generalised estimating equations. All models accounted for the repeated observations within participants (Liang and Zeger, 1986). Reported contrasts and percentages are marginal estimates on the response scale rather than raw model coefficients.

2.8.1. Average responses between study runs

The first analysis examined whether the two study runs differed in their average responses across the complete set of experimental conditions. This analysis provided a broad comparison of perceived crossing safety, post trial ratings, trigger use, and head movement between the samples.

Each model included the four experimental factors and estimated the adjusted difference between the fixed and randomised runs. The 40 conditions were given equal weight when calculating the estimated average for each run. This prevented the result from depending on the realised frequency of a condition at a particular trial position.

2.8.2. Responses across experimental conditions

The second analysis examined whether the pattern of responses across the 40 experimental conditions differed between the fixed and randomised runs. Such a difference would indicate that conclusions about a particular experimental manipulation may depend on the method used to order the trials.

Interactions between study run and vehicle behaviour, eHMI status, relative pedestrian order, and pedestrian spacing were first tested jointly. Individual conditions were compared only after this overall interaction test. Within each outcome, follow up comparisons used simultaneous 95% confidence intervals across the 40 conditions and Holm adjusted *p* values (**?**). To examine whether a finding remained robust across the additional outcomes, the condition comparison *p* values were then subjected to a second Holm correction across outcomes. Both within outcome and global adjusted *p* values are reported when they lead to different interpretations. This procedure reduced the risk of emphasising one condition solely because it produced a small unadjusted *p* value.

2.8.3. Changes with experience

The third analysis examined whether perceived crossing safety or understanding of the vehicle's intention changed as participants gained experience. Condition adjusted profiles across all 40 trial positions were plotted to describe the shape of change over the session. Formal comparisons used the three session periods defined by the optional breaks after trials 14 and 26, together with focused contrasts between the beginning and end of the session.

The focused eHMI analysis was limited to yielding trials because the yielding animation was intended to communicate that the vehicle would stop. It examined whether the estimated difference between eHMI on and eHMI off trials changed from the beginning to the end of the session. This analysis was conducted for perceived crossing unsafety and Q3, which measured understanding of the vehicle's intention.

A complementary analysis represented experience as the number of previous yielding trials in which the eHMI had been active. This measure captured direct previous encounters with the yielding eHMI rather than general progression through the experiment.

For both representations of experience, the main between run test examined whether the amount of change differed between the fixed and randomised runs. A statistically significant change within one run, accompanied by a nonsignificant change within the other, was not interpreted as evidence that the two runs differed unless the direct between run comparison was also statistically significant.

2.8.4. Sensitivity analyses

Sensitivity analyses examined whether the main conclusions changed after adjustment for age, gender, and recent virtual reality experience; exclusion of participants who shared a realised randomised sequence; use of alternative trigger thresholds; participant level bootstrap resampling; exclusion of influential participants; prior exposure models; and more flexible models of change across the session. These analyses are reported in the Results section.

An equivalence test was not performed because a scientifically justified smallest effect size of interest had not been specified before the results were examined.

3. Results

3.1. Data quality and average responses

All 100 participants completed the 40 main trials, resulting in 4,000 analysed trials. Every trial contained the complete five second analysis window with 50 valid 100 ms intervals. All participants in the fixed run completed the same sequence. For every participant, the order reconstructed from the response files matched the corresponding condition mapping. The grouped binomial Pearson dispersion exceeded 2, which confirmed that model based binomial standard errors were inappropriate. The reported uncertainty therefore uses participant clustered robust covariance and is checked below using independent participant resampling analyses.

The estimated percentage of the five second prepassage period during which participants indicated that initiating a crossing was unsafe was 40.48% (95% CI $[35.19, 45.77]$) in the randomised run and 42.82% (95% CI $[37.51, 48.12]$) in the fixed run (Figure 2). The fixed minus randomised difference was 2.34 percentage points (95% CI $[-5.16, 9.83]$, $p = .54$). The confidence interval included differences in both directions and was too wide to support a conclusion of equivalence.

Figure 2: Estimated percentage of the five second prepassage period during which participants indicated that initiating a crossing was unsafe. Estimates give equal weight to all 40 experimental conditions. Error bars represent participant clustered 95% confidence intervals.

Table 1 summarises the average responses for the other main measures. The runs did not differ reliably in mean trigger use, the percentage of trials with at least one trigger activation, the perceived influence of the other pedestrian, understanding of vehicle intention, or horizontal head orientation at vehicle passage.

Table 1
Average responses estimated with participant clustered marginal models.

Measure	Randomised	Fixed	Difference	95% CI	Family adjusted p
Time perceived as unsafe (%)	40.48	42.82	2.34	$[-5.16, 9.83]$	.54
Mean trigger value (0–1)	0.40	0.42	0.02	$[-0.05, 0.10]$	.61
Trials with trigger activation (%)	72.38	78.29	5.91	$[-3.21, 15.03]$	.61
Q1: influence of the other pedestrian	21.30	15.25	−6.05	$[-12.67, 0.57]$	.15
Q2: influence of pedestrian spacing	27.70	18.81	−8.89	$[-15.68, -2.09]$	.031
Q3: understanding of vehicle intention	68.10	65.26	−2.84	$[-9.37, 3.69]$	.39
Head orientation at passage (degrees)	−5.70	−0.84	4.86	$[-2.46, 12.19]$	.77

Differences are fixed minus randomised. The perceived unsafety comparison was the main dependent measure. The other p values use Holm correction within the relevant measure family. The Q2 comparison did not remain reliable after correction across all additional outcomes ($p = .14$).

Participants in the fixed run gave lower average ratings to Q2, which asked how much the distance between the participant and the other pedestrian affected the crossing decision. This difference remained after correction across the three rating questions but not after the more conservative correction across all additional outcomes. Q1, concerning the influence of the other pedestrian, and Q3, concerning understanding of vehicle intention, did not show reliable average differences.

Trigger use. The additional trigger measures also showed no reliable average differences. The estimated time to the first trigger active interval was 1.23 s from the start of the analysis window in the randomised run and 1.39 s in the fixed run (difference = 0.15 s, 95% CI $[-0.24, 0.54]$). Among trials with an activation, the estimated percentage in which participants subsequently returned to a safe response was 52.63% and 50.67%, respectively (difference = −1.97 percentage points, 95% CI $[-12.34, 8.41]$). The estimated time to the first return to safe was 2.83 s in the randomised run and 2.94 s in the fixed run (difference = 0.11 s, 95% CI $[-0.12, 0.33]$). Together, these results indicate that the absence of a reliable difference in time perceived as unsafe was not concealing a reliable difference in when participants first activated the trigger or returned to a safe response.

Head movement. Participants in both runs turned their heads substantially towards the approaching vehicle before it passed and were closer to a forward orientation at passage. The estimated minimum prepassage orientation was $-75.48°$ in the randomised run and $-75.67°$ in the fixed run (difference = $-0.18°$, 95% CI $[-3.20, 2.83]$). At passage, the corresponding estimates were $-5.70°$ and $-0.84°$ (difference = $4.86°$, 95% CI $[-2.46, 12.19]$). The runs also did not differ reliably in head orientation change across passage, prepassage orientation variability, or yaw activity. Figure 3 shows the participant mean distributions for the four principal head movement summaries.

Thus, no reliable overall between run difference was detected in perceived crossing unsafety, trigger use, or head movement, although the estimates remained compatible with modest differences. The next analysis examined whether the absence of reliable average differences concealed differences in the pattern of responses across particular experimental conditions.

3.2. Responses across experimental conditions

The pattern of perceived crossing unsafety across the 40 experimental conditions differed between runs ($p = .029$), although none of the 40 individual condition comparisons remained reliable after simultaneous inference. After correction across the additional outcome families, study run by condition interactions also remained for Q2 ($p = .003$), Q3 ($p = .015$), peak trigger value ($p = .011$), whether the trigger was activated ($p = .005$), minimum prepassage head orientation ($p < .001$), prepassage head orientation variability ($p < .001$), and head yaw activity ($p = .010$). These omnibus findings indicate that the shape of the response pattern across conditions differed between runs. They do not indicate that one run produced uniformly higher or lower responses, or that trial scheduling caused the differences.

The head movement findings are important in this respect. The runs had nearly identical average minimum head orientation, variability, and yaw activity, yet these measures changed differently across the 40 experimental conditions. No individual head movement condition remained reliable after simultaneous confidence intervals and multiplicity correction. It was therefore not possible to attribute the omnibus differences to one particular traffic situation. The same

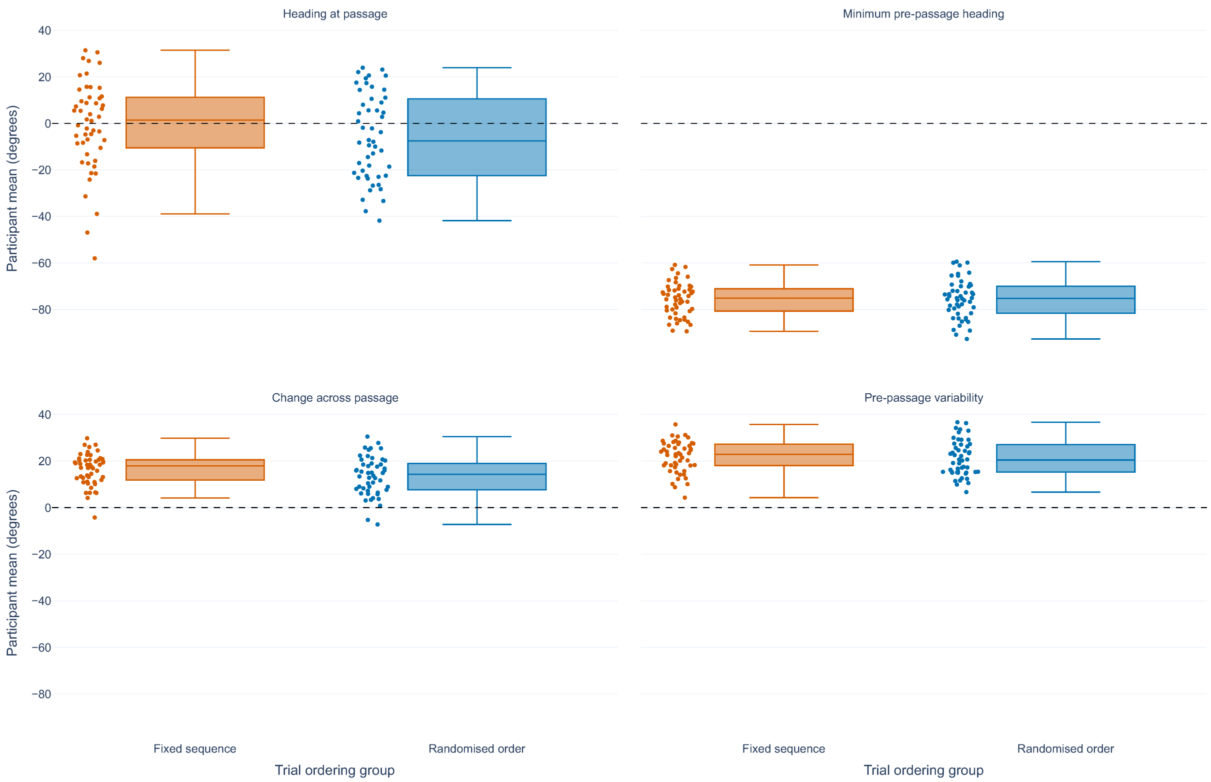

Figure 3: Participant mean head movement outcomes in the fixed and randomised runs. Negative heading values indicate leftward orientation towards the approaching vehicle. The panels show horizontal head orientation at vehicle passage, the most leftward orientation before passage, change in orientation across passage, and variability in orientation during the five second prepassage window. Each point represents one participant.

pattern applied to peak trigger value and trigger activation: the overall profiles differed, but no individual condition remained reliable after correction. The complete set of follow up comparisons was therefore summarised numerically rather than displayed as a large condition heatmap.

Only one individual condition remained reliable after correction across all condition comparisons and outcomes. Q2 asked, "How much did the distance between you and the other pedestrian affect your decision to cross the road?" The clearest difference occurred in a yielding trial in which the eHMI was off, the participant was positioned second, and pedestrian spacing was 2 m. The estimated Q2 rating was 53.41 in the randomised run and 30.08 in the fixed run. The fixed minus randomised difference was −23.34 rating points (simultaneous 95% CI [−40.93, −5.74], globally adjusted $p = .011$).

This condition was always presented as trial 1 in the fixed run. In the randomised run, it occurred at 29 different positions across participants, spanning trials 1 to 40. The difference may therefore reflect how participants initially interpreted or used the spacing scale, a difference between the two study samples, or both. Because the condition and trial position were identical for every participant in the fixed run, these explanations cannot be separated.

Table 2 presents the two largest Q2 differences for yielding trials with the eHMI off and the participant positioned second. The related 4 m condition was always trial 3 in the fixed run. Its confidence interval excluded zero and its adjusted p value was .034 within Q2, but the comparison did not remain reliable after correction across all conditions and outcomes ($p = .52$).

These findings do not show that fixed ordering systematically increased or decreased responses. Instead, the absence of a reliable average difference coexisted with different condition specific response patterns. A local condition difference is difficult to interpret when the condition is always presented at one stage of the experiment and trial scheduling is confounded with study run.

Table 2
Q2 comparisons for yielding trials with the eHMI off and the participant positioned second.

Spacing (fixed trial)	Randomised	Fixed	Difference	Simultaneous 95% CI	Adjusted p (Q2; global)
2 m (1)	53.41	30.08	−23.34	[−40.93, −5.74]	.00075; .011
4 m (3)	45.02	26.64	−18.39	[−36.20, −0.57]	.034; .52

Differences are fixed minus randomised rating points. Confidence intervals and the first adjusted p value use simultaneous inference across the 40 Q2 conditions. The second p value applies the additional Holm correction across condition comparisons and outcomes.

3.3. Changes with experience

Figure 4 provides a descriptive overview of the condition adjusted responses across all 40 trial positions. Perceived crossing unsafety, mean trigger value, and the probability of trigger activation changed little in the adjusted profiles, although the fixed run showed higher estimated values towards the end of the session. Q3, which measured understanding of vehicle intention, generally increased across the session. The head movement profiles also changed over trial position, particularly around the two optional breaks. These curves are descriptive summaries with substantial uncertainty. Local rises, falls, or differences at a particular trial position were therefore not treated as inferential evidence of change.

For a more interpretable formal comparison, Figure 5 summarises condition adjusted responses across the three session segments defined by the optional breaks. Perceived crossing unsafety and the probability of trigger activation did not show reliable early to late changes in either run. The estimated Q3 rating of vehicle intention understanding increased from the first to the final segment by 4.35 points in the randomised run (95% CI [0.39, 8.30]) and by 5.68 points in the fixed run (95% CI [1.14, 10.23]). Neither change remained reliable after correction across all analysed outcomes, and the difference in early to late change between runs was also uncertain (1.34 points, 95% CI [−4.68, 7.36], $p = .66$).

The clearest session segment change concerned prepassage head yaw activity. It decreased by 3.77°/s from the first to the final segment in the randomised run (95% CI [−5.52, −2.02], globally adjusted $p < .001$). The fixed run showed a smaller estimated decrease of 2.40°/s (95% CI [−4.83, 0.03]). However, the difference in change between runs was not reliable (1.37°/s, 95% CI [−1.62, 4.37], $p = .37$). Prepassage head orientation variability also decreased in the randomised run by 2.53° (95% CI [−4.25, −0.81], globally adjusted $p = .050$), but its change likewise did not differ reliably between runs. These results suggest some reduction in head movement over the session, but not a clear difference between the ordering methods.

Experience with the yielding eHMI. The focused experience analysis examined whether the eHMI on versus off difference changed within yielding trials. Q3 asked participants to rate how well they understood the intention of the vehicle. In the randomised run, the estimated eHMI effect on Q3 increased by 17.81 rating points between trial 1 and trial 40 (95% CI [8.59, 27.04], adjusted $p = .002$). The estimated increase in the fixed run was 11.85 points, with a wide confidence interval that included no change (95% CI [−13.92, 37.62], $p = .37$).

The direct comparison between runs is the relevant test of whether the development of eHMI understanding differed between ordering methods. This comparison was not reliable: the fixed minus randomised difference in change was −5.96 points (95% CI [−33.23, 21.30], $p = .67$).

The analysis based on previous relevant encounters produced the same interpretation. Between zero and nine previous yielding trials with the eHMI active, the estimated eHMI effect on Q3 increased by 13.71 points in the randomised run (95% CI [5.03, 22.38], adjusted $p = .022$). The increase was 6.49 points in the fixed run (95% CI [−6.27, 19.24]), and the difference in change between runs was not reliable (−7.22 points, 95% CI [−22.65, 8.21], $p = .36$).

No comparable pattern was observed for perceived crossing unsafety. Changes in the eHMI effect on perceived unsafety were small and uncertain for both trial position and previous yielding eHMI encounters (Figure 6). The results therefore indicate that the estimated contribution of the eHMI to vehicle intention understanding became larger with experience in the randomised run, but they do not show that this development differed between the two runs.

Broader prior exposure analyses. In addition to the focused yielding eHMI analysis, exploratory models examined whether responses depended on the number of previous encounters with the current level of vehicle behaviour, eHMI status, pedestrian distance, or relative pedestrian order. Several family adjusted tests involved previous exposure to pedestrian distance levels, including perceived unsafety, Q1, Q2, and mean trigger value. However, these

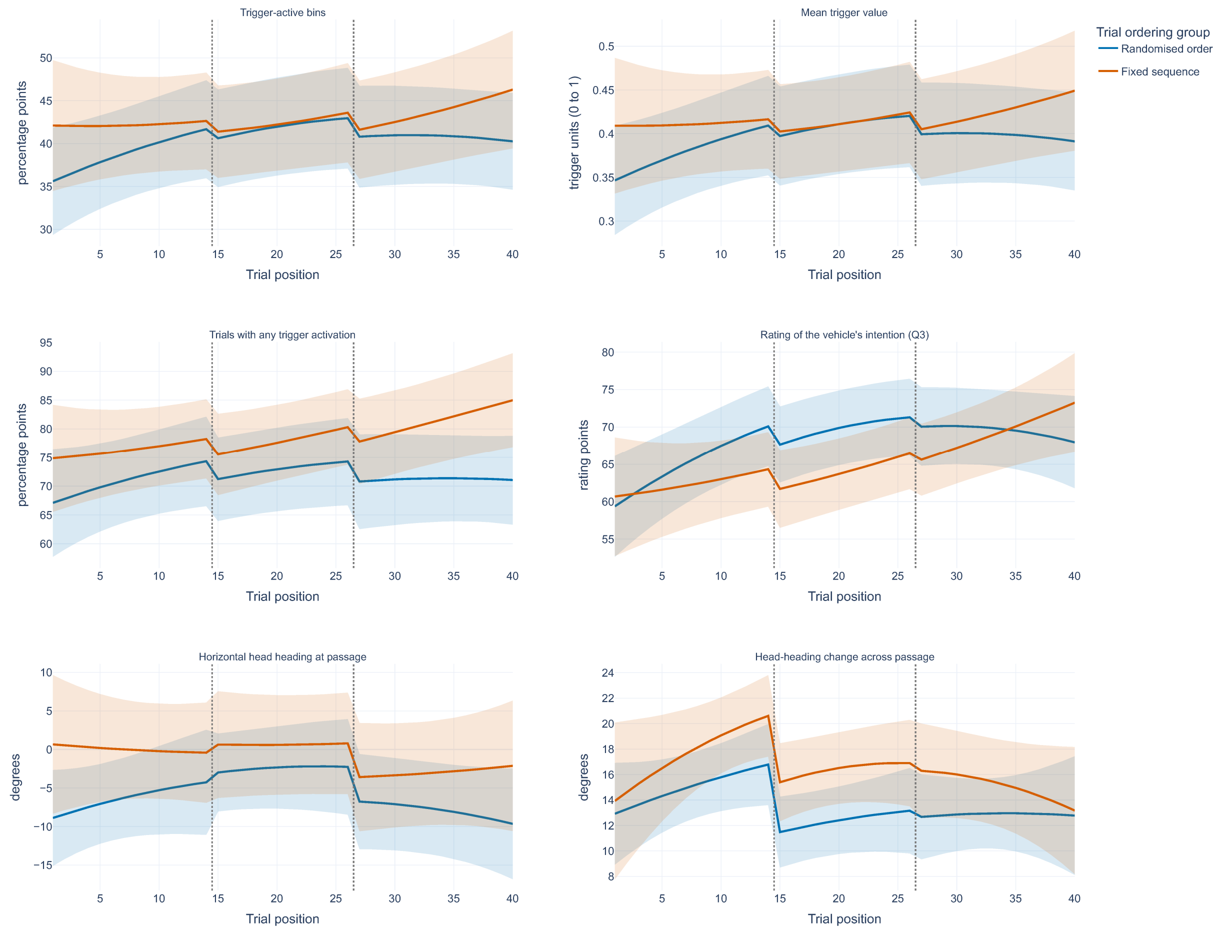

Figure 4: Condition adjusted response profiles across the 40 realised trial positions. Panels show perceived crossing unsafety, mean trigger value, the probability of any trigger activation, Q3 understanding of vehicle intention, horizontal head orientation at vehicle passage, and change in head orientation across passage. Vertical dotted lines mark the optional breaks after trials 14 and 26. Lines represent model adjusted estimates and shaded bands represent participant clustered 95% confidence intervals. The profiles are descriptive; formal temporal conclusions were based on the session segment and focused experience contrasts.

findings did not remain reliable under the correction across all outcomes and were not consistent across related behavioural measures. They therefore indicate possible sequence specific exposure associations rather than a coherent general learning effect.

Flexible temporal analyses. Flexible spline models checked whether the temporal conclusions depended on linear or segment based representations of trial position. After correction across all outcomes, the group by spline trajectory tests were not reliable for perceived crossing unsafety (adjusted $p = .42$), Q3 understanding of vehicle intention ($p = 1.00$), prepassage head orientation variability ($p = .42$), or yaw activity ($p = 1.00$). A quadratic model identified a difference between runs in the shape of prepassage orientation variability, but its joint temporal test was uncertain and the pattern was not reproduced by the session segment or spline analyses. The temporal evidence therefore does not support a general claim that one ordering method produced stronger familiarisation or behavioural adaptation.

Sensitivity analyses. The primary perceived unsafety point estimate was consistent across the planned checks. The participant cluster bootstrap estimate was 2.34 percentage points with a 95% percentile interval of $[-5.14, 9.74]$. The Bayesian participant bootstrap produced a posterior mean difference of 2.35 percentage points and a 95% credible interval of $[-4.86, 9.58]$. Demographic adjustment gave a difference of 1.80 percentage points (95% CI $[-6.13, 9.73]$).

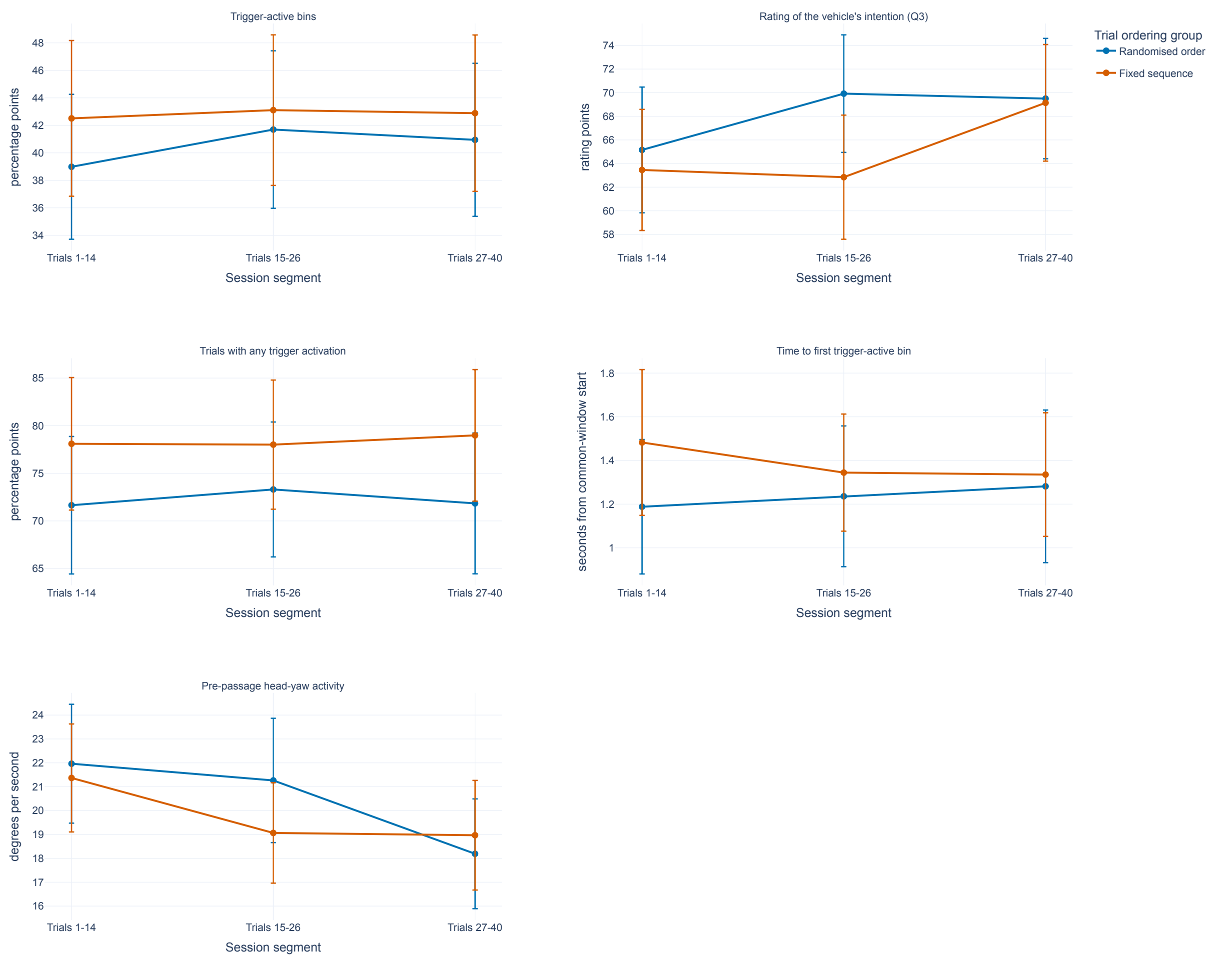

Figure 5: Condition adjusted estimates across the three session segments defined by the optional breaks. Panels show perceived crossing unsafety, Q3 understanding of vehicle intention, the probability of any trigger activation, time to the first trigger active interval, and prepassage head yaw activity. Points represent estimated means and error bars represent participant clustered 95% confidence intervals.

Thresholds of 0.05 and 0.50 produced differences of 2.43 and 2.13 percentage points, respectively. Excluding the two participants who shared a randomised sequence produced a difference of 2.41 percentage points. Across the 100 leave one participant out analyses, the estimate ranged from 1.52 to 3.20 percentage points and never changed sign. These checks support the consistency of the estimated direction and magnitude, while retaining substantial uncertainty about the true between run difference.

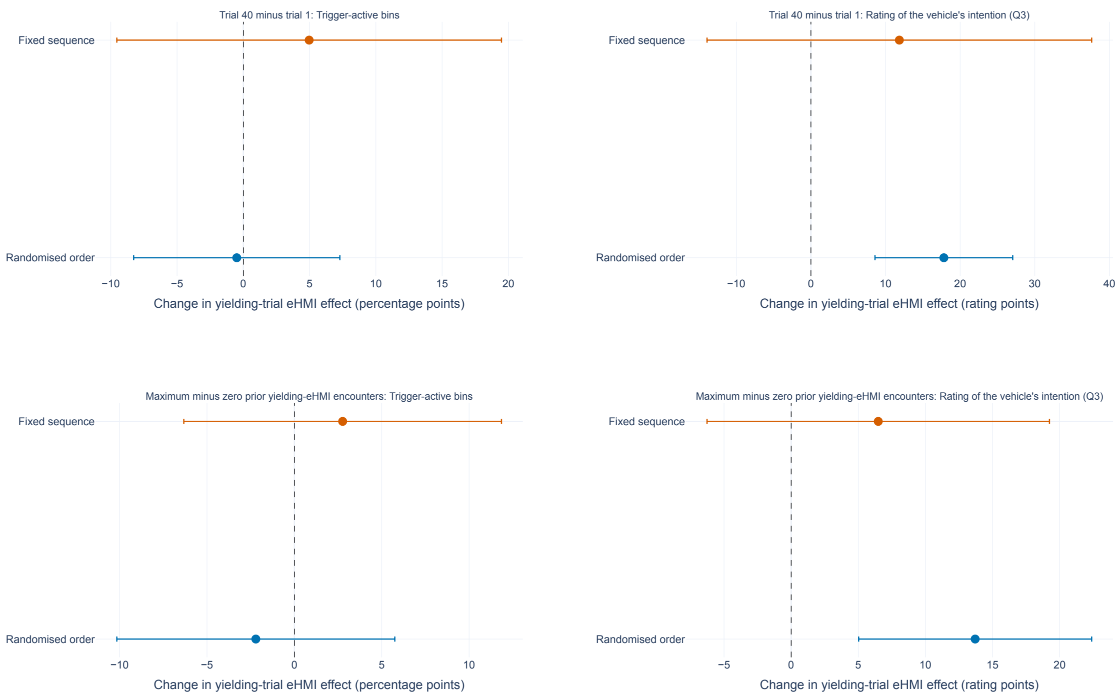

Figure 6: Changes in the estimated eHMI on versus off effect within yielding trials. The upper panels compare trial 40 with trial 1. The lower panels compare nine with zero previous yielding trials in which the eHMI was active. Q3 measured understanding of the vehicle's intention. Points represent estimated changes and error bars represent 95% confidence intervals.

4. Discussion

4.1. Principal findings

This study examined whether conclusions from a repeated exposure virtual reality pedestrian experiment were consistent across two independent runs that used one fixed sequence or participant specific randomised sequences. Four findings are central. First, no reliable overall between run difference was detected in perceived crossing unsafety, trigger use, or head movement, although the confidence intervals did not establish equivalence. Second, the pattern of responses across the 40 experimental conditions differed for several ratings, trigger measures, and head movement measures despite the absence of reliable average differences. Third, the clearest individual condition difference concerned the perceived influence of pedestrian spacing in a condition that was always presented first in the fixed run. Fourth, experience related changes were present in reported vehicle intention understanding and in head movement, but the amount of change did not differ reliably between runs. The focused eHMI contrast increased with experience in the randomised run, while the session segment analysis showed decreasing yaw activity, yet neither pattern established a between run difference.

The results therefore do not provide evidence that one run produced a general upward or downward shift in participants' responses. Instead, they show that an uncertain average difference can coexist with different response patterns across conditions and with changes over the session whose interpretation depends on how conditions were scheduled. This distinction is important because the scientific interpretation of a repeated measures experiment often concerns particular situations or changes with experience rather than only the grand mean. Because trial scheduling was completely associated with study run, the observed differences may reflect scheduling, differences between samples or study periods, or a combination of these influences.

4.2. Condition specific ratings, trigger use, and head movement

The average perceived crossing unsafety difference was small relative to its uncertainty, and the additional trigger measures led to the same broad conclusion. No reliable between run difference was detected in the proportion of trials with trigger activation, the timing of the initial unsafe response, or the probability of returning to a safe response after activation. These findings do not establish equivalence, but they provide no evidence of a large general difference in controller use between the samples.

The condition level analyses were more informative. The profiles for trigger activation and peak trigger value differed across conditions, although no individual trigger condition remained reliable after correction. This means that the relationship between the traffic conditions and trigger behaviour was not identical in the two runs, but the available data did not identify one specific situation that accounted for the difference. Treating only the average trigger response as the result would therefore miss part of the ordering comparison.

The head movement results showed the same distinction between average behaviour and condition specific behaviour. In both runs, participants turned markedly towards the approaching vehicle before passage and were closer to facing forward when the vehicle passed. No reliable overall between run difference was detected in minimum orientation, orientation at passage, prepassage variability, or yaw activity. This pattern is compatible with a broadly comparable amount of head checking, while the confidence intervals retain uncertainty about modest differences.

However, minimum prepassage orientation, prepassage orientation variability, and yaw activity showed different patterns across the 40 experimental conditions. These omnibus differences remained after correction across the analysed outcomes, despite the absence of a reliable average difference. The condition response structure therefore differed between study runs for several head movement measures. No individual head movement condition survived the simultaneous follow up comparisons, so the data do not justify attributing the difference to one specific condition or to trial scheduling alone.

Head movement also has a limited psychological meaning. Turning towards the vehicle is compatible with checking its approach, but head mounted display orientation does not identify where the eyes were directed or what information participants attended to. The results should therefore be described as differences in head orientation and movement rather than differences in visual attention.

The clearest individual condition result concerned Q2, which measured the perceived influence of pedestrian spacing. The yielding, eHMI off, participant second condition at 2 m produced lower ratings in the fixed run and was always the first fixed trial. Participants may need initial trials to understand which aspects of the scene are relevant and to establish how they use a 0–100 rating scale. The related 4 m condition occurred at trial 3 and showed a difference in the same direction, although it did not remain reliable after correction across all outcomes.

This interpretation remains uncertain because the two runs used independent participant samples. The Q2 difference may reflect early scale use, an unmeasured difference between samples, or both. The methodological point is that the fixed sequence makes these explanations inseparable: the condition, its position, and its preceding experience were the same for every fixed run participant. Similar overall means therefore do not make trial order irrelevant, while a local difference in a fixed sequence cannot automatically be attributed to the manipulated condition.

4.3. Changes in eHMI understanding and head movement with experience

The condition adjusted trial profiles provide a descriptive account of how responses developed across all 40 trials. They show little estimated change in perceived crossing unsafety and trigger activation, alongside a general increase in Q3 understanding and changes in head movement. However, the profiles also contain local fluctuations and wide confidence bands, so their visual shape should not be interpreted as evidence that the runs differed at particular trial positions. The session segment analysis provides the more interpretable formal summary. Perceived crossing unsafety and trigger activation did not show reliable early to late changes, whereas Q3 understanding increased numerically from the first to the final segment in both runs. The direct difference in Q3 change between runs was uncertain. This pattern is compatible with developing reported understanding during the session, but it does not indicate that one run produced a stronger general increase.

The focused eHMI analysis asked a more specific psychological question: whether the difference in understanding between eHMI on and eHMI off yielding trials became larger with experience. This contrast increased with trial position and with previous relevant eHMI encounters in the randomised run. The convergence is consistent with evidence that pedestrians may revise their interpretation of external vehicle communication through repeated encounters (Lee et al., 2022; Colley et al., 2022, 2024). Broader recent evidence likewise links repeated automated vehicle exposure to changes in trust, crossing timing, visual sampling, and psychophysiological arousal (Hübner et al., 2025; Shahi et al., 2026).

The result should be interpreted as a change in reported understanding rather than as direct evidence that every participant learned the signal. A larger Q3 contrast could reflect improved interpretation, increased confidence, or greater reliance on the eHMI. These possibilities are psychologically different, particularly because repeated exposure can also encourage inappropriate reliance on an external signal (Kaleefathullah et al., 2022).

More importantly, the change in the Q3 eHMI contrast did not differ reliably between the fixed and randomised runs. A significant change in one run and a nonsignificant change in the other is not evidence of a difference between runs. The data support an experience related increase in the eHMI contribution to reported vehicle understanding within the randomised run, but not the stronger conclusion that participant specific randomisation produced faster or greater learning.

Perceived crossing unsafety did not show the same experience related eHMI pattern. Understanding what the vehicle intends and judging whether it is safe to cross are related but not interchangeable. The safety judgement also depends on vehicle movement, distance, the other pedestrian, and the participant's decision criterion. An eHMI may therefore become easier to interpret without producing a corresponding change in the duration for which crossing is considered unsafe.

The session segment analysis also showed decreasing prepassage yaw activity in the randomised run and a smaller decrease in the fixed run. Prepassage orientation variability decreased in the randomised run as well. These changes are compatible with general familiarisation: participants may have needed fewer or smaller head movements once the task and vehicle trajectories became familiar. However, the direct early to late changes did not differ reliably between runs. Head movement should also not be equated with gaze or attention. The defensible conclusion is that some head movement measures changed during the session, not that randomisation altered visual information sampling.

The additional temporal analyses qualify this interpretation. Event defined trigger models showed that the absence of a reliable overall difference in trigger use was not concealing clear differences in response timing, and flexible trajectory models did not reproduce a consistent between run difference after correction. Local rises, falls, or apparent curve separation should therefore not be interpreted as evidence of a general ordering effect unless supported by the direct between run tests.

4.4. Implications for repeated exposure experiments

The value of participant specific randomisation depends on the research question. In the present data, one fixed sequence and participant specific randomised sequences yielded comparable point estimates for several averages, but the uncertainty was too large to establish equivalence. More importantly, this correspondence did not extend to the complete response structure across conditions. Ratings, trigger activation, and head movement all provided examples in which condition response profiles differed despite the absence of reliable average differences.

When researchers are interested in learning, familiarisation, or changing interpretation, a fixed sequence creates a further problem. Every condition occurs at one position and after one specific history. A temporal change may consequently reflect the changing composition of conditions rather than a psychological change with experience. Conversely, a condition effect may partly reflect how experienced participants were when that condition occurred.

Participant specific randomisation reduces these associations by distributing conditions across positions and preceding experiences. It does not eliminate learning or familiarisation. Rather, it makes those processes less dependent on a particular condition. When unrestricted randomisation would create undesirable repetitions or implausible transitions, restricted randomisation, Williams designs, or Latin square arrangements may offer more suitable alternatives (Williams, 1949; Jones and Kenward, 2003; Brooks, 2012).

Proposed schedules should be inspected before data collection for balance across early, middle, and late trials and for the distribution of relevant preceding exposures. The realised order should then be stored for every participant and linked to the response data. This is especially important when technical interruptions, repeated trials, or implementation details can make the realised order differ from the intended schedule.

4.5. Limitations and future study

The main limitation is that participants were not assigned concurrently at random to the fixed and randomised scheduling methods. Trials were shuffled separately within the randomised run, but the fixed run was conducted earlier. Trial scheduling was therefore inseparable from study run, sample, and time of data collection. Differences in recruitment, participant expectations, laboratory context, or experimenter practice may have contributed to the observed contrasts. Adjustment for age, gender, and recent virtual reality experience did not change the conclusions, but adjustment for measured characteristics cannot remove unmeasured differences between study runs. The empirical contrasts should consequently not be interpreted as causal effects of fixed versus randomised ordering.

Only one fixed sequence was examined. The placement of the largest Q2 differences at trials 1 and 3 may be specific to this sequence. The condition specific trigger and head movement profiles may also differ under another fixed schedule. This limitation is part of the methodological concern: reusing one sequence can make results depend on arbitrary scheduling decisions. A stronger prospective study would assign participants concurrently to participant specific randomisation and several balanced fixed or counterbalanced sequences.

The sample did not provide precise evidence about modest between run differences. The 95% confidence interval for the main perceived unsafety comparison ranged from approximately −5.16 to 9.83 percentage points. Absence of statistical significance is therefore not evidence of equality. A future equivalence study would require an independently justified smallest effect size of interest and a sample size selected for that purpose.

The findings are specific to the present virtual environment, vehicle behaviour, stationary co-pedestrian, and conditional light based eHMI. Participants indicated whether initiating a crossing felt unsafe but did not physically cross. Different response patterns may emerge with moving pedestrian groups, multiple vehicles, inconsistent eHMIs, alternative communication modalities, more complex traffic, or longer sessions.

Finally, the controller and head mounted display measures capture only parts of pedestrian behaviour. The trigger threshold and five second window were examined in sensitivity analyses, but the measure does not represent crossing initiation or body movement. Head orientation provides information about gross movement towards and away from the vehicle, but it does not measure eye gaze or visual attention. Future studies combining crossing behaviour, head movement, and eye tracking could clarify whether condition specific head movement differences reflect information seeking, uncertainty, or other aspects of pedestrian strategy.

5. Conclusion

No reliable overall between run difference was detected in perceived crossing unsafety, trigger use, or head movement, although the confidence intervals did not establish equivalence. The pattern of responses across the 40 conditions nevertheless differed for several ratings, trigger measures, and head movement measures. The strongest individual condition difference concerned a spacing rating presented at the first fixed trial, so the condition response could not be separated from early session experience or differences between samples. Reported understanding of vehicle intention and head movement changed over the session, but the amount of change did not differ reliably between runs. Because trial scheduling was completely associated with study run, these findings do not identify a causal effect of ordering method. They instead demonstrate the interpretive limitation of a fixed sequence: condition, trial position, and preceding experience cannot be separated. Participant specific randomisation or balanced counterbalancing is therefore important when repeated exposure research concerns condition specific behaviour or psychological change across a session.

Data availability

In line with current open science practices and recommendations for transparency in automotive user research (Ebel, Bazilinskyy, Colley, Goodridge, Hock, Janssen, Sandhaus, Srinivasan and Wintersberger, 2024), the authors openly provide these research artefacts to support reproducibility, collaboration and further advancements in the field. The materials used in the study, analysis code and anonymised responses of the participants are available at https://www.dropbox.com/scl/fo/4pphss4roc9stcue0904o/AN76AaVvkSZHDkR4rW9_xvs?rlkey=sfnc6k4neajka793zxlw845wx. The maintained versions of the analysis code and the VR environment are available at https://github.com/Shaadalam9/multiped-learning

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